

Solutions

1. **Solution:** $r = OP = \sqrt{(5-2)^2 + (1-(-3))^2} = \sqrt{3^2 + 4^2} = 5$. So $C = 2\pi r = 10\pi$, $A = \pi r^2 = 25\pi$.

2. **Solution:** $\ell = AB = 9 - 1 = 8$, $w = BC = |-4 - 2| = 6$. $P = 2(8 + 6) = 28$, $A = 8 \cdot 6 = 48$.

3. **Solution:** $PQ = 6$, $PR = \sqrt{(6-0)^2 + (8-0)^2} = 10$, $QR = 8$. Since $PQ^2 + QR^2 = 6^2 + 8^2 = 36 + 64 = 100 = PR^2$, right at Q . $P = 6 + 8 + 10 = 24$, $A = \frac{1}{2} \cdot 6 \cdot 8 = 24$.

4. **Solution:** $AB = 6$, $AC = \sqrt{(0+3)^2 + (5-0)^2} = \sqrt{9+25} = \sqrt{34}$, $BC = \sqrt{(0-3)^2 + (5-0)^2} = \sqrt{9+25} = \sqrt{34}$. Perimeter $= 6 + 2\sqrt{34}$. Area with base $AB = 6$ and height $= 5$ gives $A = \frac{1}{2} \cdot 6 \cdot 5 = 15$.

5. **Solution:** Center is midpoint of E, F : $((-4+6)/2, (2+2)/2) = (1, 2)$. Diameter $= |6 - (-4)| = 10$, so $r = 5$, $C = 10\pi$, $A = 25\pi$.

6. **Solution:** Diagonal $d = \sqrt{(6-(-2))^2 + (11-3)^2} = \sqrt{8^2 + 8^2} = 8\sqrt{2}$. Square side $s = \frac{d}{\sqrt{2}} = 8$. Perimeter $= 32$, area $= 64$.

7. **Solution:** Reflect across x-axis: $(x, y) \mapsto (x, -y)$ so $A'(3, -1)$, $B'(7, -4)$, $C'(1, -5)$. Perimeter and area are unchanged (rigid motion).

8. **Solution:** Translate first: $P' = (-2 + 5, 3 - 4) = (3, -1)$. Then 90° CCW: $(x, y) \mapsto (-y, x)$, so $P'' = (1, 3)$.

9. **Solution:** Rotate first: 90° CCW sends $P(-2, 3) \mapsto (-3, -2)$. Then translate $(+5, -4)$ to $P'' = (2, -6)$. Not the same as Q8 $\Rightarrow$ order matters.

10. **Solution:** Reflection across x-axis sends $(x, y) \mapsto (x, -y)$. If $A' = (4, -7)$, then $A = (4, 7)$.

11. **Solution:** (a, b) satisfies $(-3 + a, 2 + b) = (5, -1) \Rightarrow a = 8, b = -3$. Apply to $N(0, 4)$: $N' = (0 + 8, 4 - 3) = (8, 1)$.

12. **Solution:** Test formulas: 90° CCW $(x, y) \mapsto (-y, x)$ gives $(-(-1), 4) = (1, 4) \neq (-1, -4)$. 180° gives $(-4, 1) \neq (-1, -4)$. 270° CCW $(x, y) \mapsto (y, -x)$ yields $(-1, -4)$, so 270° CCW.

13. **Solution:** Base = 8 (horizontal), height = $5 - 2 = 3$. $A = \frac{1}{2} \cdot 8 \cdot 3 = 12$. Rigid motions preserve area, so the rotated triangle also has area 12.

14. **Solution:** $RS = \sqrt{(9 - 2)^2 + (4 - 1)^2} = \sqrt{49 + 9} = \sqrt{58}$; $ST = \sqrt{(5 - 9)^2 + (-2 - 4)^2} = \sqrt{16 + 36} = \sqrt{52} = 2\sqrt{13}$; $TR = \sqrt{(2 - 5)^2 + (1 - (-2))^2} = \sqrt{9 + 9} = 3\sqrt{2}$.
Perimeter = $\sqrt{58} + 2\sqrt{13} + 3\sqrt{2}$.

15. **Solution:** (a) $D = (0, 6)$. (b) $A = 8 \cdot 6 = 48$, $P = 2(8 + 6) = 28$. (c) Reflection across x-axis preserves lengths $\Rightarrow$ area and perimeter unchanged.

16. **Solution:** $r = \sqrt{12^2 + 5^2} = \sqrt{144 + 25} = 13$. $C = 26\pi$, $A = 169\pi$.

17. **Solution:** Translate: $A' = (1 - 3, 2 + 1) = (-2, 3)$, $B' = (5 - 3, 2 + 1) = (2, 3)$, $C' = (5 - 3, 6 + 1) = (2, 7)$. Then 270° CCW $(x, y) \mapsto (y, -x)$: $A'' = (3, 2)$, $B'' = (3, -2)$, $C'' = (7, -2)$.

18. **Solution:** 180° CCW $(x, y) \mapsto (-x, -y) = (7, -9)$. So original was $(-7, 9)$.

19. **Solution:** Base $\overline{XY}$ from -4 to 4 has length 8 . Height is the vertical distance from Z to $y = 0$: $|7 - 0| = 7$. $A = \frac{1}{2} \cdot 8 \cdot 7 = 28$. Perimeter: $XZ = \sqrt{(1 + 4)^2 + (7 - 0)^2} = \sqrt{25 + 49} = \sqrt{74}$, $YZ = \sqrt{(1 - 4)^2 + (7 - 0)^2} = \sqrt{9 + 49} = \sqrt{58}$, plus $XY = 8$. So $P = 8 + \sqrt{74} + \sqrt{58}$.

20. **Solution:** Diagonal of inscribed square $= 2r = 20$. Side $s = \frac{20}{\sqrt{2}} = 10\sqrt{2}$. Area $= s^2 = 200$. Translation preserves distances $\Rightarrow$ side and area unchanged.